

Identifying and Mitigating Distributed Denial of Service Attacks

A distributed denial of service (DDoS) attack involves the paralysing of a network by flooding it with data from several individual sources, to the detriment of genuine users. The identification and mitigation of such attacks is an important issue for systems administrators.

With the advent and adoption of technology-loaded devices, a number of challenges are regularly resolved by cyber security professionals.

These include breach of privacy, data sniffing and theft, integrity, access control, fake traffic, channel damage, and many others. Cyber security experts are continuously working to enforce and integrate a higher level of security in the devices as well as the network infrastructure so that users can access and use the technology without any vulnerability issues.

In a technology based environment, the hacking community regularly launches numerous attacks. A number of algorithms and mechanisms have been devised to cope with such virtual attacks. Research is still under way in the domain of securing applications, devices, networks and computing infrastructure.

Broadly, there are two types of attacks in any network based environment.

Passive attacks: In any network environment, when there is an attempt at sniffing the data channels to copy secret information, it is considered to be a passive attack. In such cases, the modification of files, directories or credentials is not done. Passive attacks are primarily used to monitor the remote system or server, to spy on private information.

Active attacks: In the case of active attacks, the effect and damage of the assault is instant and, often, the victim machine would not have been prepared for such attacks. These include injection of malicious traffic to damage the remote system, updating or deleting remote files, modifying authentication files and much more.

Distributed denial of service (DDoS) attack

A distributed denial of service (DDoS) attack is a powerful assault, in the taxonomy of active attacks, which is used to

restrict access to services from authenticated users. In DDoS attacks, genuine users are not allowed to use a system or service because of excessive traffic.

In very simple terms, fake or malicious traffic is generated in large volumes on a server in order to overload it, thus

resulting in the network getting choked or jammed. DDoS attacks are often known as jamming attacks because fake traffic or data packets overload the server delivering a particular service. Due to this attack, other users are not allowed to access the service because of massive congestion in the network.

As an example, let's suppose there is a limit of 200 concurrent users

who can access a website. In a DDoS attack, 200 sources of website access can be generated. After those 200 connections are captured by fake traffic, the genuine users will not be able to access that particular website because of traffic overload or channel congestions.



Types of DDoS attacks

- **Application layer based DDoS:** Such an attack is used to target and damage the application layer of the network. The effect of this attack is measured in terms of requests per second (RPS). A large number of RPS are generated, which becomes a load for the network.
- **Protocol based DDoS:** In this type of attack, the resources and related modules of the server are the victims. Bandwidth is not captured in protocol based DDoS.
- **Volume based DDoS:** Bandwidth is the key target here: it is saturated and flooded with massive traffic in volume based DDoS attacks. If such an attack is successful, the server crashes and major flaws occur.

HULK based DDoS

Whenever there is a DDoS attack on a website, it is known